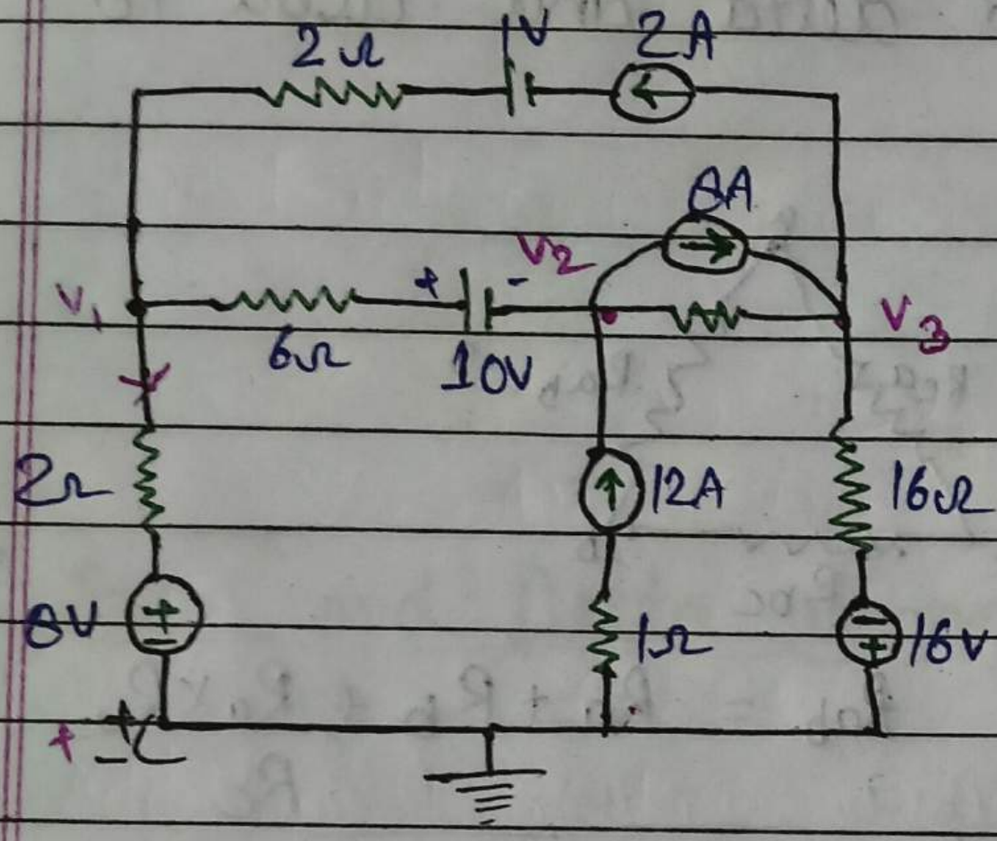
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⊕ Nodel Analysis :→

→ It is based on kirchoff current law and Ohm's law.

→ It is applicable on both plane and non-planner

Ques→



Node(1) →

$V_1 > V_2$ and $V_1 > V_3$

$$2 = \frac{V_1 - 8 - 0}{2} + \frac{V_1 - 10 - V_2}{6} \Rightarrow$$

Node(2) → $V_2 > V_1$ and $V_2 > V_3$

$$N \to 12 = 8 + \frac{V_2 + 10 - V_1}{6} + \frac{V_2 - V_3}{8}$$

$$-4V_1 + 7V_2 - 3V_3 = 56 \longrightarrow (II)$$

Node(3) →

$V_3 > V_1$ and $V_3 > V_2$

$$4V_1 - V_2 = 46 \longrightarrow (I)$$

$$8 = 2 + \frac{V_3 + 16 - 0}{16} + \frac{V_3 - V_2}{8} \longrightarrow (III)$$

$$6 = \frac{V_3 + 16}{16} + \frac{V_3 - V_2}{8} \Rightarrow 96 = (V_3 + 16) + 2(V_3 - V_2)$$

$$96 = -2V_2 + 3V_3 + 16$$

$$-2V_2 + 3V_3 = 80 \longrightarrow (3)$$

eq (3) + (4) to eliminate V_3 →

$$(-2V_2 + 3V_3) + (6V_2 - 3V_3) = 80 + 102$$

$$V_2 = \frac{182}{4} = 45$$

$$-2V_2 + 3V_3 = 80$$

$$-2(45.5) + 3V_3 = 80$$

$$-91 + 3V_3 = 80$$

$$\left[V_3 = \frac{171}{3} = 57V\right]$$

$$4V_1 - V_2 = 46$$

$$4V_1 - 45.5 = 46$$

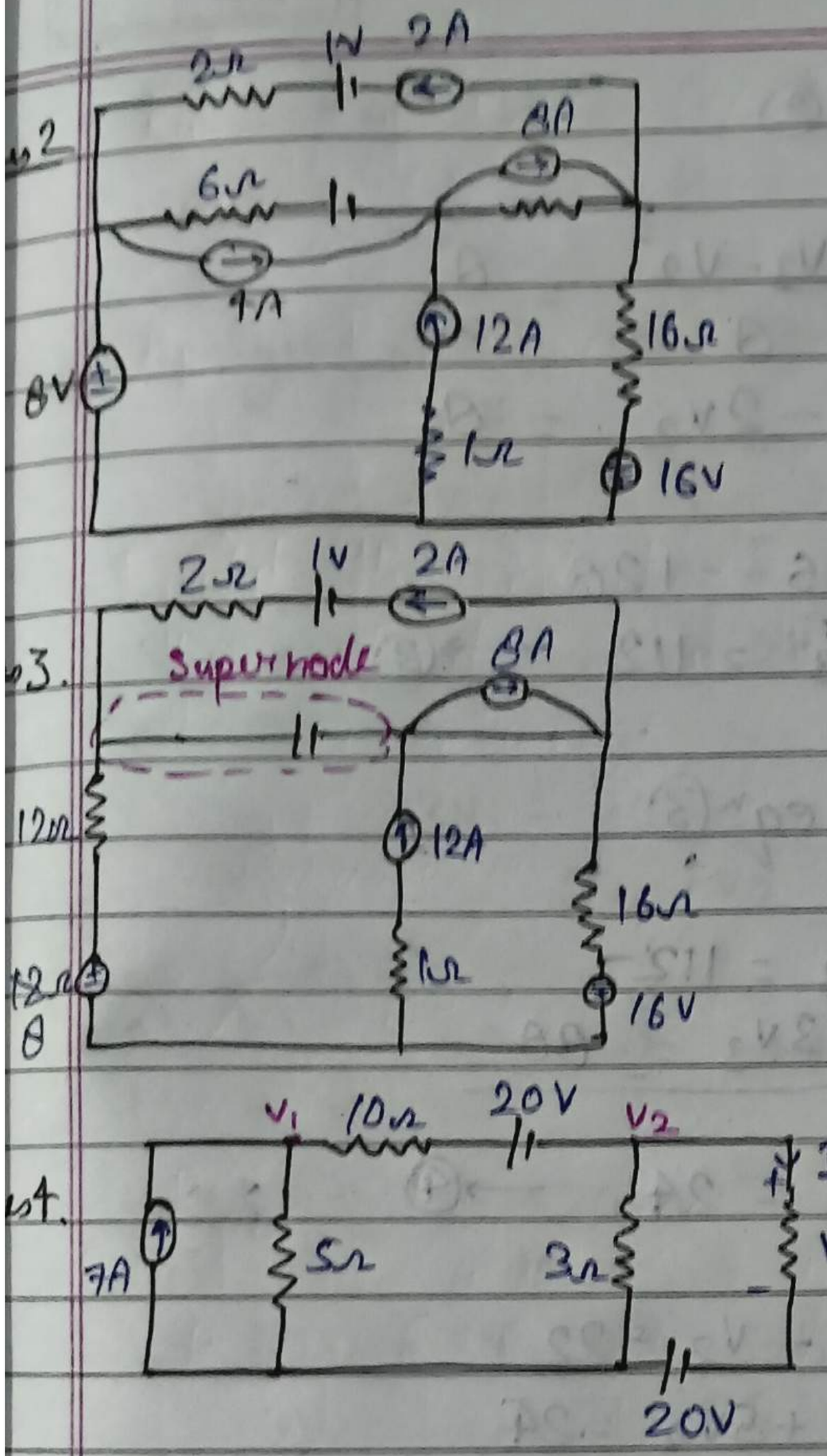
$$4V_1 = 46 + 45.5$$

$$V_1 = \frac{91.5}{4} = 22.875$$

$[V_1 = 22.875]$

$V_1 = 8V$

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2- Node 1 :

Applying KCL at node 1

$$\frac{V_1 - 8 - 0}{2} + 4 + \frac{V_1 - 10 - V_2}{6} = 2A$$

$$\frac{3V_1 - 24 + 24 + V_1 - 10 - V_2}{6} = 2$$

$$4V_1 - V_2 - 10 = 12$$

$$4V_1 - V_2 = 22 \longrightarrow (1)$$

Applying KCL at node ②

$$\frac{V_3 - 16 - 0}{16} + 2 + \frac{V_3 - V_2}{8} = 0$$

$$\frac{V_3 - 16 + 32 + 2V_3 - 2V_2}{16} = 0$$

$$-2V_2 + 3V_3 + 16 = 128$$

$$-2V_2 + 3V_3 = 112 \longrightarrow ③$$

Subtract eq② from eqⁿ③

$$-2V_2 + 3V_3 = 112$$

$$\underline{4V_1 - 7V_2 + 3V_3 = 88}$$

$$-4V_1 + 5V_2 = 24 \longrightarrow ④$$

Add eqⁿ ① and ④

$$4V_1 - V_2 = 22$$

$$\underline{-4V_1 + 5V_2 = 24}$$

$$4V_2 = 46$$

$$V_2 = \frac{46}{4} \qquad [V_2 = 11.5V]$$

from eq①

$$4V_1 = 22 + V_2$$

$$4V_1 = 22 + 11.5$$

$$V_1 = \frac{33.5}{4}$$

$$\boxed{V_1 = 8.375\ V}$$

from eq③ → $3V_3 - 2V_2 = 112 \Rightarrow 3V_3 = 112 + 2V_2$

$$3V_3 = 112 + 2 \times 11.5$$

$$3V_3 = 112 + 23 = 135$$

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$$\left[V_3 = \frac{135}{3} = 45V\right] =$$

Solve: Applying KCL at node ①

$$\frac{V_1 - 20 - V_2}{10} + \frac{V_1 - 0}{5} = 7$$

$$\frac{V_1 - 20 - V_2 + 2V_1}{10} = 7$$

$$3V_1 - 20 - V_2 = 70$$

$$3V_1 - V_2 = 90 \longrightarrow ①$$

Applying KCL at node → ②

$$\frac{V_2 - 0}{3} + \frac{V_2 + 20 - V_1}{10} + \frac{V_2 + 20}{2} = 0$$

$$\frac{10V_2 + 3V_2 + 60 - 3V_1 + 15V_2 + 300}{30} = 0$$

$$-3V_1 + 28V_2 = -360 \longrightarrow ②$$

Adding eq ① and eq ②, we get →

$$+3V_1 - V_2 = 90$$
$$-3V_1 + 28V_2 = -360$$
$$27V_2 = -270$$

$$V_2 = -10V$$

from eqn →

$$3V_1 = 90 + V_2$$
$$90 - 10 = 80$$
$$[V_1 = 26.66V]$$

Using ohm's law →

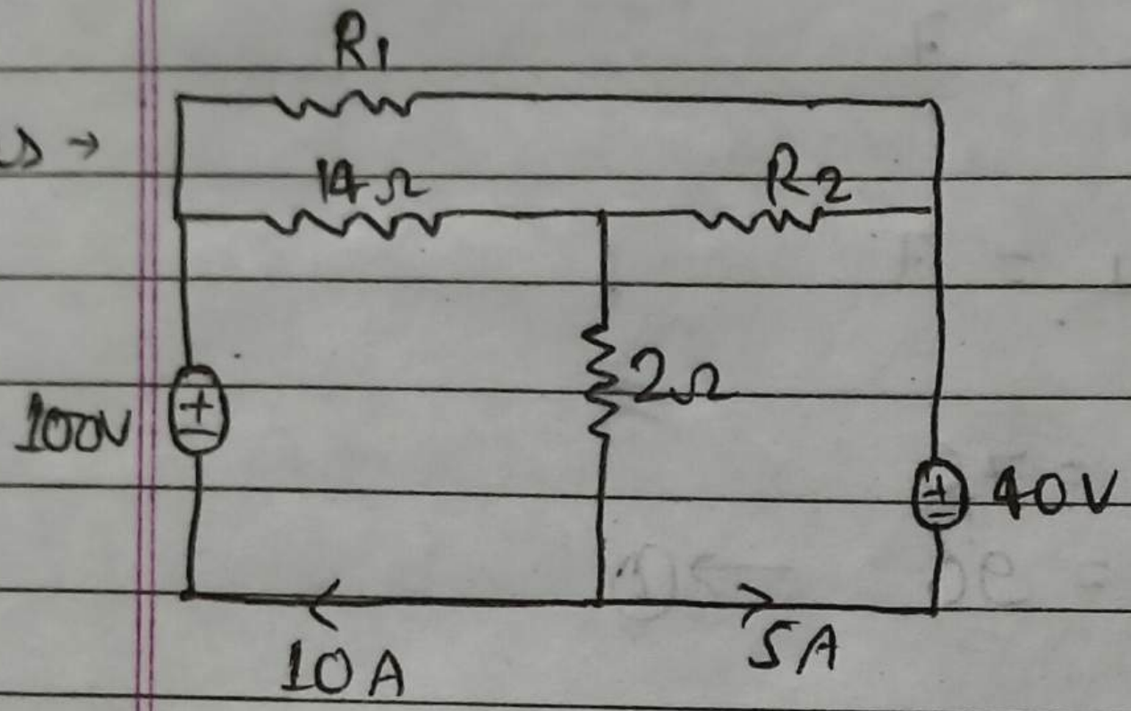
$$I = \frac{V}{R} \quad , \quad I_x = \frac{V_2 + 20 - 0}{2} = \frac{-10 + 20}{2}$$

$I_x = 5A$

$V_x = I_x \times 2$

$V_x = 5 \times 2 = 10V$

Ques →



Applying KCL and KVL

Loop ① $100 - V - 30 = 0$

$V = 70$

$I = \frac{70}{14} = 5A$

Loop ②–

$40 - V_1 - 30 = 0$

$V_1 = 10$

$10R_2 = 10$

$[R_2 = 1\Omega]$

loop ③

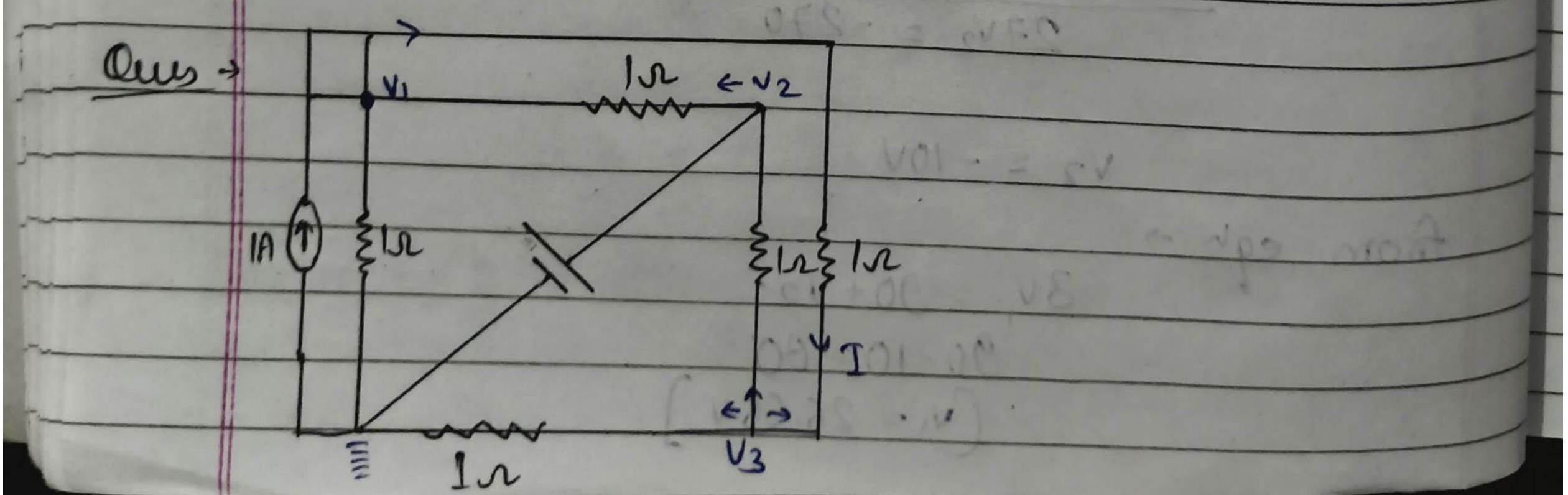
$V - 5R_1 - V_1 = 0$

$70 - 5R_1 - 10 = 0$

$5R_1 = 60$

$[R_1 = 12\Omega]$

Ques →



$V_2 - 0 = 1$

$[V_2 = 1V]$

Supernode →

$$\Rightarrow \frac{V_2 - V_1}{1} + \frac{V_2 - V_3}{1} + \frac{1 - 0 - V_3}{1} + \frac{0 - V_1}{1} = 0$$

$$\Rightarrow V_2 - V_1 + V_2 - V_3 + 1 - V_3 - V_1 = 0$$

$$-2V_1 + 2V_2 - V_3 + 1 - V_3 - V_1 = 0$$

$$-2V_1 + 2 - 3V_3 = -1$$

$$-2V_1 - 2V_3 = -3$$

$$[2V_1 + 2V_3 = 3] \rightarrow (1)$$

node → ①

$$\frac{V_1 - 0}{1} + \frac{V_1 - V_2}{1} + \frac{V_1 - V_3}{1} = 1$$

$$3V_1 - V_2 - V_3 = 1$$

$$3V_1 - 1 - V_3 = 1$$

$$3V_1 - V_3 = 2 \rightarrow (2)$$

node → ③

$$\frac{V_3 - 0}{1} + \frac{V_3 - V_2}{1} + \frac{V_3 - V_1}{1} = 0$$

$$-V_1 - V_2 + 3V_3 = 0$$

$$-V_1 + 3V_3 = 1 \rightarrow (3)$$

from eqn ① and ②

$$-2V_1 + 6V_3 = 2$$

$$2V_1 + 2V_3 = 3$$

$$8V_3 = 5 \qquad \left[V_3 = \frac{5}{8}\right]$$

from eqn ③

$$V_1 = 3V_3 - 1$$

$$V_1 = \frac{3 \times 5}{8} - 1 = \frac{15}{8} - 1 = \frac{7}{8}$$

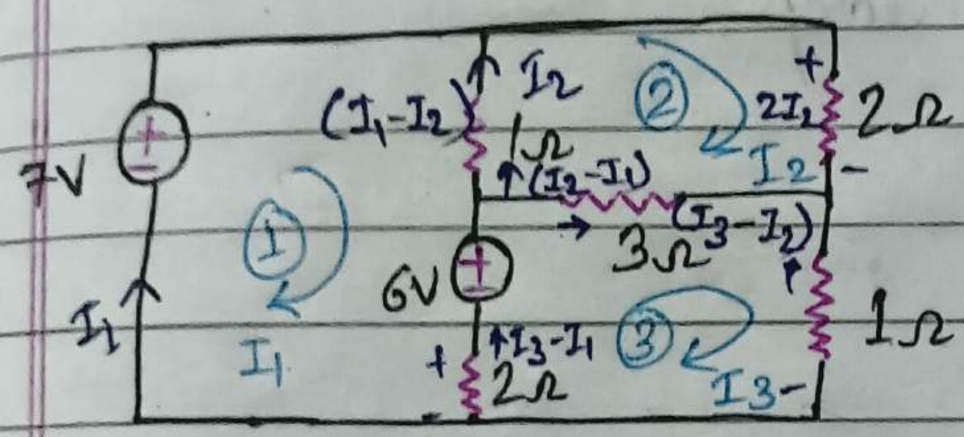
$$I = \frac{V_1 - V_3}{1} = \frac{7/8 - 5/8}{1} = \frac{2}{8} = \frac{1}{4}$$

$$[I = 0.25A]$$

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(#) Loop/mesh Analysis →

i) It is applicable on (loop) planner.
ii) It is based on KVL and Ohm's law.



→ loop ① → $I_1 > I_2$ and $I_1 > I_3$

$$7 - 1(I_1 - I_2) - 6 - 2(I_1 - I_2) = 0 \longrightarrow ①$$
$$3I_1 - I_2 - 2I_3 = 1 \longrightarrow ①$$

→ Loop ② → $I_2 > I_1$ and $I_2 > I_3$

$$-2I_2 - 3(I_2 - I_3) - 1(I_2 - I_1) = 0 \longrightarrow ②$$
$$I_1 - 6I_2 + 3I_3 = 0 \longrightarrow ②$$

→ Loop ③ $I_3 > I_1$ and $I_3 > I_2$

$$6 - 3(I_3 - I_2) - 1I_3 - 2(I_3 - I_1) = 0 \longrightarrow ③$$
$$2I_1 + 3I_2 - 6I_3 = -6 \longrightarrow ③$$

from eqn ② → input eq^n ①

$I_1 = 6I_2 - 3I_3$

$3(6I_2 - 3I_3) - I_2 - 2I_3 = 1$

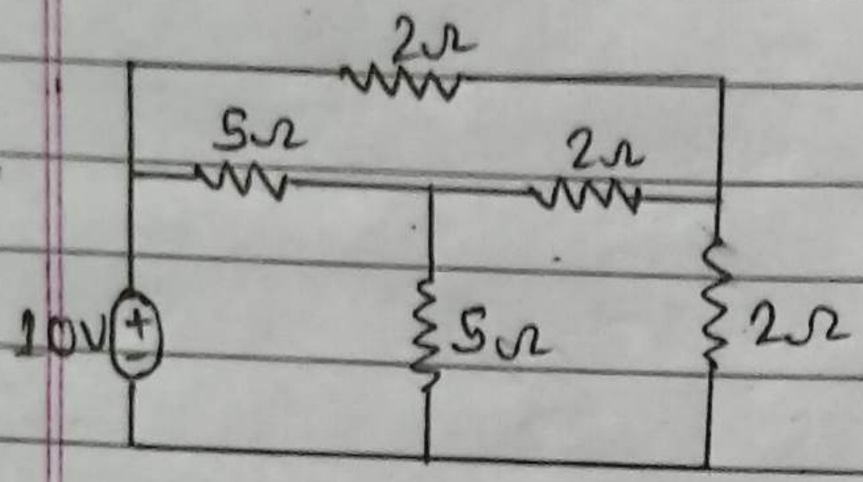
$17I_2 - 11I_3 = 1 \longrightarrow ④$

$5(2) - 4I_3 = -2$
$10 - 4I_3 = -2$
$-4I_3 = -2 - 10$
$-4I_3 = -12$
$I_2 = 2A$

$68I_2 - 44I_3 - 55I_2 - 44I_3$
$= 4 - (-22)$
$= 68I_2 - 55I_2 - 88I_3 = 26$
$I_3 = 3A$

$I_1 = 6I_2 - 3I_3$
$I_1 = 6(2) - 3(3)$
$I_1 = 12 - 9 = 3.$
$I_1 = 3A$

Ques→ find current in each loop.



loop ①

Consider ($I_1 > I_2$ and $I_1 > I_3$)

applying KVL →

$$10 - 5(I_1 - I_3) - 5(I_1 - I_2) = 0$$

$$10 - 10I_1 + 5I_2 + 5I_3 = 0$$

$$10 = 10I_1 - 5I_2 - 5I_3$$

$$2I_1 - I_2 - I_3 = 2 \rightarrow ①$$

loop ②

applying KVL

$$-5(I_2 - I_1) - 2(I_2 - I_3) - 2I_2 = 0$$

$$5I_1 - 9I_2 + 2I_3 = 0 \rightarrow ②$$

loop ③

applying KVL

$$-5(I_3 - I_1) - 2I_3 - 2(I_3 - I_2) = 0$$

$$5I_1 + 2I_2 - 8I_3 = 0 \rightarrow ③$$

eq^n ② and ③

$$5I_1 - 9I_2 + 2I_3 = 0$$

$$\underline{5I_1 + 2I_2 - 8I_3 = 0} \qquad \left(I_3 = \frac{11I_2}{10} \right)$$

$$-11I_2 + 10I_3 = 0$$

$5 \times eq① - 2 \times eq②$

$$10I_1 - 5I_2 - 5I_3 = 0$$
$$\underline{10I_1 - 18I_2 + 4I_3 = 0}$$

$$13I_2 - 9I_3 = 10$$
$$13I_2 - \frac{99I_2}{10} = 10 \quad \because I_3 = \frac{11I_2}{10}$$

Now →

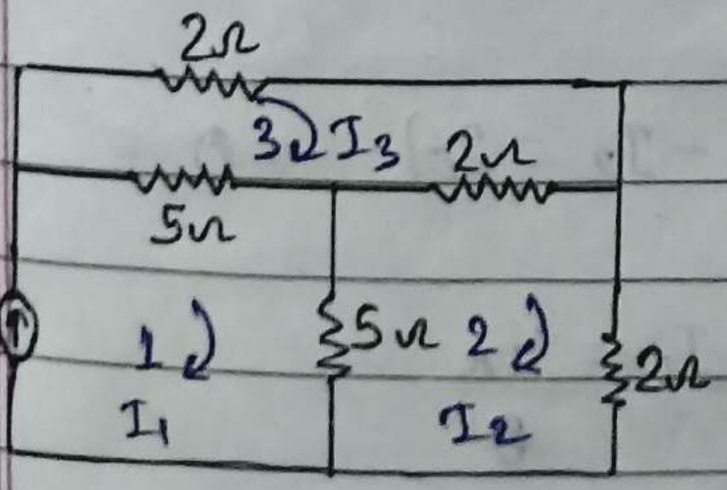
$$I_2(130-99) = 100$$
$$\left[I_2 = \frac{100}{31} A\right]$$

$$I_3 = \frac{11I_2}{10} = \frac{11}{10} \times \frac{100}{31} = \frac{110}{310} \text{ Amp}$$

from eq① →

$$2I_1 = I_2 + I_3 + 2$$
$$= \frac{100}{31} + \frac{110}{31} + 2 = \frac{210+62}{31} = \frac{272}{31}$$
$$\left[I_1 = \frac{136}{31} \text{ Amp}\right]$$

Find current in each loop →



only I_1 current flowing in loop 1

$[I_1 = 10 \text{Amp}]$

loop ② applying KVL

$$-5(I_2 - I_1) - 2(I_2 - I_3) - 2I_2 = 0$$
$$-5(I_2 - 10) - 2I_2 + 2I_3 - 2I_2 = 0$$
$$50 = 9I_2 - 2I_3 \rightarrow ①$$

loop ③ → applying KVL

$$-2I_3 - 2(I_3 - I_2) - 5(I_3 - I_1) = 0$$
$$-9I_3 + 2I_2 + 5I_1 = 0$$
$$5 \times 10 = -2I_2 + 9I_3$$
$$-2I_2 + 9I_3 = 50 \rightarrow ②$$

from eqⁿ ② and ③

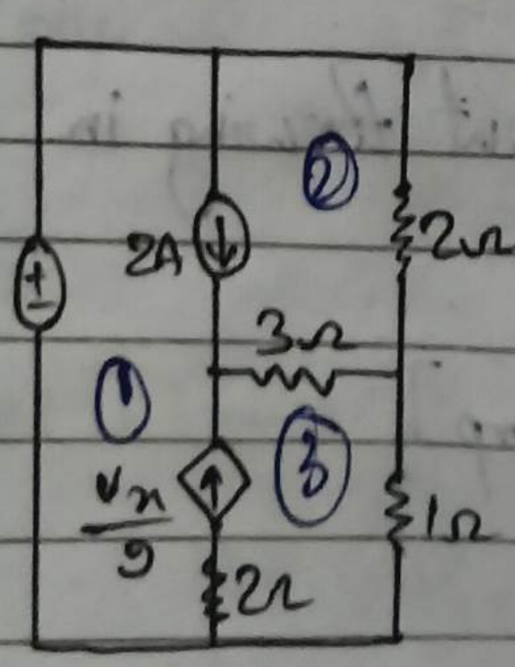
$$9I_2 - 2I_3 = 50$$
$$+\ 2I_2 + 9I_3 = 50$$
$$11I_2 - 11I_3 = 0$$
$$[I_2 = I_3]$$

from eqⁿ ① and ②

$$9I_2 - 2I_3 = 50$$
$$-2I_2 + 9I_3 = 50$$
$$7I_2 + 7I_3 = 100$$
$$I_2 + I_3 = \frac{100}{7} \rightarrow ③$$

$$\left[I_2 = \frac{50}{7}\ A \ , \ I_3 = \frac{50}{7}\ A\right]$$

Ques →



loop ① → $I_1 - I_2 = 2A \rightarrow ①$

loop ③ $I_3 - I_1 = \frac{V_x}{9}$

and $V_x = 3(I_3 - I_2)$

$$I_3 - I_1 = \frac{3(I_3 - I_2)}{9_3}$$

$$3I_3 - 3I_1 = I_3 - I_2$$

$$2I_3 - 3(I_2 + 2) + I_2 = 0$$

$$2I_3 - 2I_2 = 6$$

$$I_3 - I_2 = 3 \longrightarrow (2)$$

using eq (3) in eq (2)

$$15 - 2I_2 - I_2 = 3$$

$$12 = 3I_2 \Rightarrow [I_2 = 4A]$$

$$I_3 = 15 - 2I_2$$

$$[I_3 = 15 - 2 \times 4 = 7A]$$

from eq (1)

$$[I_1 = 2 + I_2 \therefore I_1 = 2 + 4 = 6 \text{Amp}]$$

no of nodal eqn = (n-1)

no of eqn in mean analysis = b - a

AC fundamentals

Average value →

When a dc source is connected across the circuit, it transfer certain amount of charge in a given time period, same circuit is connected across the AC supply to adjust the AC is such a way that it transfer same no. of charges in same time period as in the case of dc. The A.C value is known as avg. value →

$$f(t) = \frac{1}{T}\int_0^T f(t)\,dt$$

(or)

$$\text{avg} = \frac{\text{area}}{\text{time}}$$

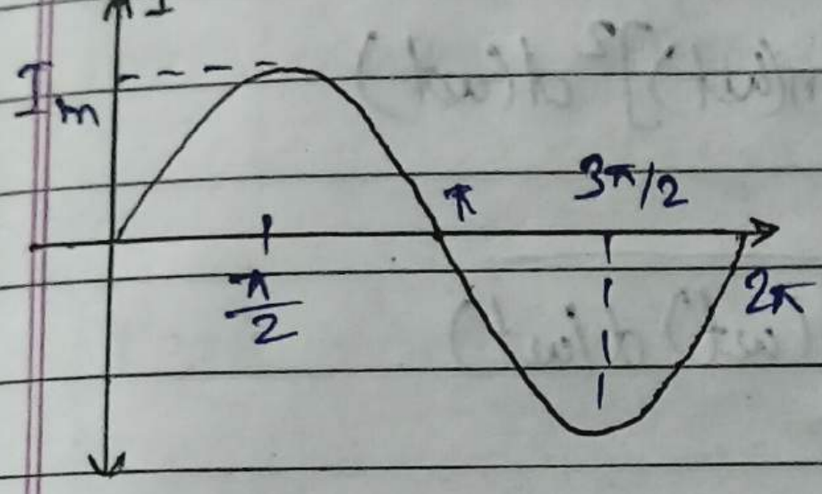
Root mean square value or effective value →

When a resistance it is connected across a (DC) supply, it produce sudden amount of heat in given time (t) some resistance is connected the (AC) supply to adjust the (AC) value in such way that it produce same same amount of heat in some time period as in the case of D.C.

The A.C value is known as rms value.

$$f(f)_{rms} = \sqrt{\frac{1}{T}\int_0^T [f(t)]^2 dt}$$

1) Sinusoidal Wave form →



$[I = I_m \sin\omega t]$

2) Triangular Wave form →

→ To find average value →

$$f(t)\text{-}avg = \frac{1}{T}\int_0^T f(t)\,dt$$

$$I_{avg} = \frac{1}{\pi}\int_0^{\pi} I_m \sin\omega t\, d(\omega t) = \frac{I_m}{\pi}\int_0^{\pi} \sin(\omega t)\, d(\omega t)$$

$$I_{avg} = \frac{I_m}{\pi}[-\cos\omega t]_0^{\pi}$$

$$I_{avg} = \frac{I_m}{\pi}\left[-\cos\pi - (-\cos 0^\circ)\right]$$

$$= \frac{I_m}{\pi}[-(-1) - (-1)]$$

$$\left[I_{avg} = \frac{2I_m}{\pi} = 0.637\, I_m\right]$$

To find rms value →

$$f(t)_{rms} = \sqrt{\frac{1}{T}\int_0^T [f(t)]^2 dt}$$

$$I_{rms} = \sqrt{\frac{1}{\pi}\int_0^{\pi} [I_m \sin(\omega t)]^2 d(\omega t)}$$

$$I_{rms} = \sqrt{\frac{I^2_m}{\pi}\int_0^{\pi} \sin^2(\omega t)\, d(\omega t)}$$

$$I_{rms} = \sqrt{\frac{I_m^2}{\pi} \times \frac{1}{2}\int_0^{\pi} (1 - \cos\omega t)\, d(\omega t)}$$

$$I_{rms} = \sqrt{\frac{I_m^2}{2\pi}\left[\omega t - \frac{\sin 2\omega t}{2}\right]_0^{\pi}}$$

$$I_{rms} = \sqrt{\left(\frac{I_m}{\sqrt{2\pi}}\right)^2\left[\omega t - \frac{\sin 2\omega t}{2}\right]_0^{\pi}}$$

$$I_{rms} = \frac{I_m}{\sqrt{2\pi}}\sqrt{[(\pi - 0) - (0 - 0)]}$$

$$\boxed{I_{rms} = \frac{I_m}{\sqrt{2}} = 0.707\, I_m}$$

form factor → It is ratio of rms value to avg value

KN

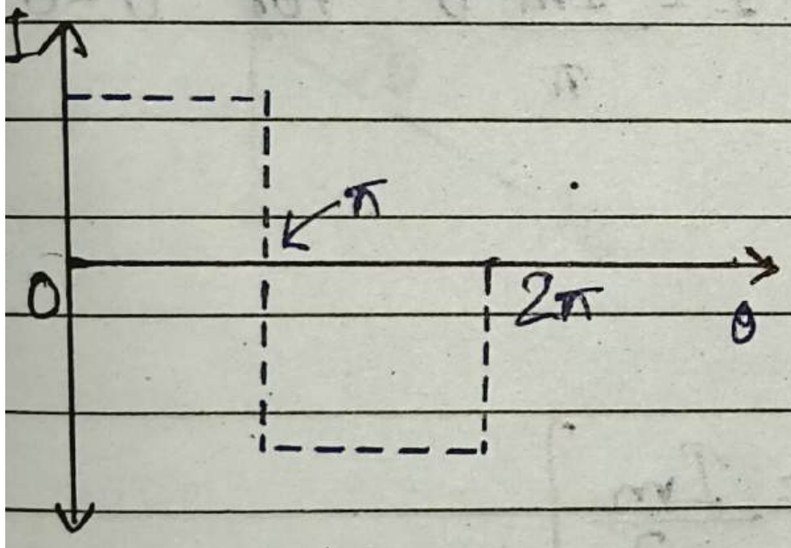
Peak factor →

It is the ratio of peak value to rms value for Sinusoidal wave form →

$$\text{form factor} = \frac{\text{rms value}}{\text{avg value}} = \frac{I_m/\sqrt{2}}{2I_m/\pi} = \frac{\pi}{2\sqrt{2}}$$

$$\text{Peak factor} = \frac{\text{peak value}}{\text{rms value}} = \frac{I_m}{I_m/\sqrt{2}} = \sqrt{2}$$

Square wave form →



$I = I_m$ for $0 < \theta \leq \pi$

Average value →

$$I_{avg} = \frac{1}{\pi}\int_0^{\pi} I(\theta)\,d\theta = \frac{1}{\pi}\int_0^{\pi} I_m\,d\theta = \frac{I_m}{\pi}[\theta]_0^{\pi}$$

$$I_{avg} = \frac{I_m}{\pi} \times \pi$$

$$\boxed{I_{avg} = I_m}$$

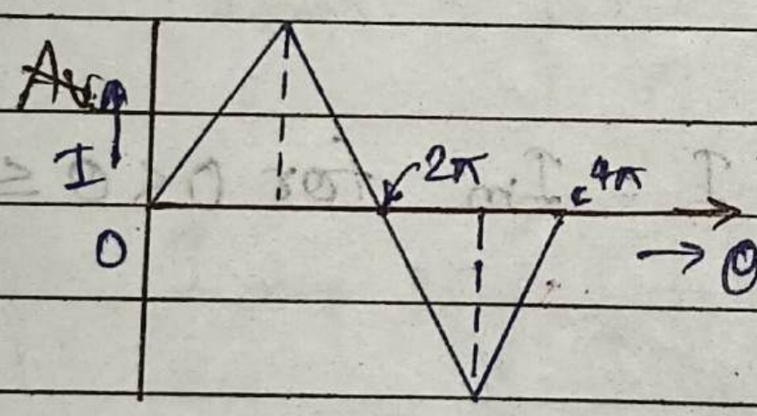
RMS value →

$$I_{rms} = \sqrt{\frac{1}{T}\int_0^{\pi}[I(\theta)]^2 d\theta} = \sqrt{\frac{I_m}{\pi}\int_0^{\pi}\theta\, d\theta}$$

$$= \sqrt{\frac{I_m^2}{\pi} \times \pi}$$

$$\boxed{I_{rms} = I_m}$$

I Triangular Waveform →



$$I = \frac{I_m}{\pi}\theta \quad \text{for } 0 < \theta < 2\pi$$

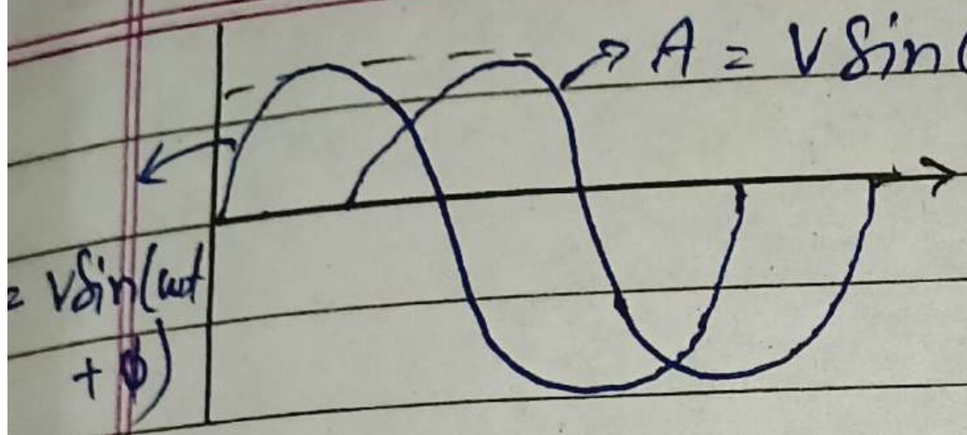
Average value → $\boxed{I_{avg} = \frac{I_m}{2}}$

RMS value →

$$\boxed{I_{rms} = \frac{I_m}{\sqrt{3}}}$$

→ Phasor Diagram ←

The phasor diagram is the graphical representation of wave. It is a complex number which represents amplitude and phase of sinusoidal waves.



→ B leads A (or) A lags B

